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JEE MAIN · QUESTION PAPER

JEE Main 2025 (Online) 28th January Evening Shift

JEE Main Question Paper with Answer Key

28 Jan 2025	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	25 questions
Mathematics	25 questions
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Answer key	at the end

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Chemistry 25 questions

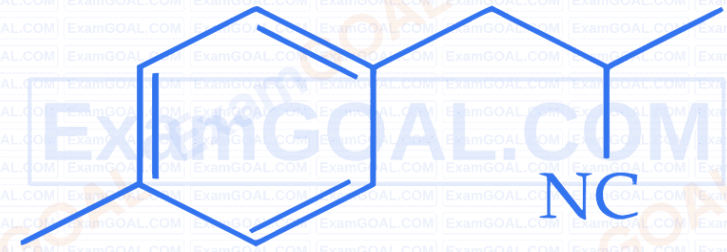
Q1. MCQ (Single Correct Answer)

Open solution

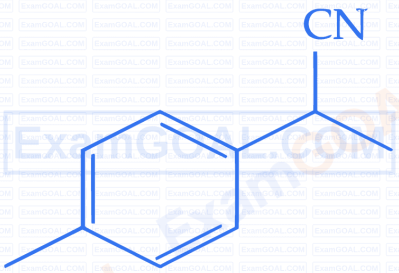
The product B formed in the following reaction sequence is:



(A)



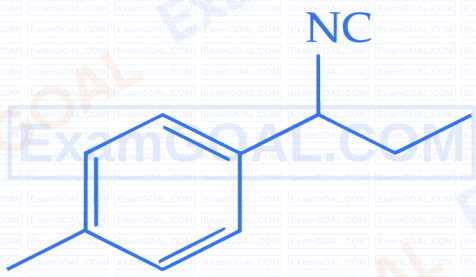
(B)



(C)



(D)



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Q2. MCQ (Single Correct Answer)

Open solution

The major product of the following reaction is:



(A) 6-Phenylhepta-3,5-diene

(B) 6-Phenylhepta-2,4-diene

(C) 2-Phenylhepta-2,4-diene

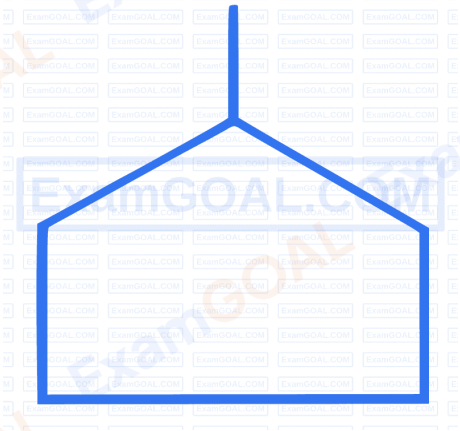
(D) 2-Phenylhepta-2,5-diene

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Q3. MCQ (Single Correct Answer)

Open solution

Given below are two statements:

Statement I:  and  are isomeric compounds.

Statement II:  and  are functional group isomers.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

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Q4. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation of central metal ion)
(A) $[CoF_6]^{3-}$	(I) d^2sp^3
(B) $[NiCl_4]^{2-}$	(II) sp^3
(C) $[Co(NH_3)_6]^{3+}$	(III) sp^3d^2
(D) $[Ni(CN)_4]^{2-}$	(IV) dsp^2

Choose the correct answer from the options given below:

(A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(C) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

(D) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Q5. MCQ (Single Correct Answer)

Open solution

Arrange the following in increasing order of solubility product :

$Ca(OH)_2, AgBr, PbS, HgS$

(A) $PbS < HgS < Ca(OH)_2 < AgBr$

(B) $HgS < AgBr < PbS < Ca(OH)_2$

(C) $HgS < PbS < AgBr < Ca(OH)_2$

(D) $Ca(OH)_2 < AgBr < HgS < PbS$

Q6. MCQ (Single Correct Answer)

Open solution

Match List-I with List-II.

List - I (Saccharides)	List - II (Glycosidic-linkages found)
(A) Sucrose	(I) α 1 – 4
(B) Maltose	(II) α 1 – 4 and α 1 – 6
(C) Lactose	(III) α 1 – β 2
(D) Amylopectin	(IV) β 1 – 4

Choose the **correct** answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

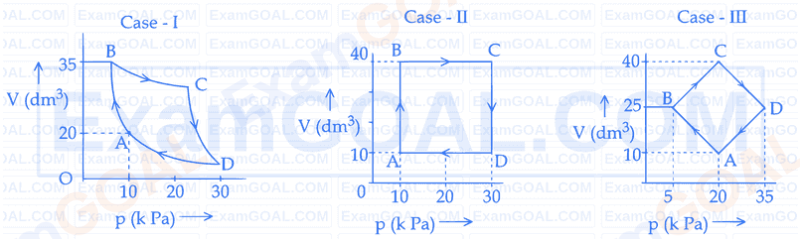
(B) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(C) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

(D) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

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Q7. MCQ (Single Correct Answer)



Open solution

An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A → B → C → D → A as shown in the three cases above.

Choose the **correct** option regarding ΔU :

(A) $\Delta U(\text{ Case-III }) > \Delta U(\text{ Case-II }) > \Delta U(\text{ Case-I })$

(B) $\Delta U(\text{ Case-I }) > \Delta U(\text{ Case-II }) > \Delta U(\text{ Case-III })$

(C) $\Delta U(\text{ Case-I }) = \Delta U(\text{ Case-II }) = \Delta U(\text{ Case-III })$

(D) $\Delta U(\text{ Case-I }) > \Delta U(\text{ Case-III }) > \Delta U(\text{ Case-II })$

Q8. MCQ (Single Correct Answer)

Open solution

Which of the following is/are not correct with respect to energy of atomic orbitals of hydrogen atom?

(A) 1s < 2p < 3d < 4s

(B) 1s < 2s = 2p < 3s = 3p

(C) 1s < 2s < 2p < 3s < 3p

(D) 1s < 2s < 4s < 3d

Choose the **correct** answer from the options given below :

(A) (A) and (B) only **(B)** (A) and (C) only **(C)** (B) and (D) only **(D)** (C) and (D) only

Q9. MCQ (Single Correct Answer)

Open solution

The amphoteric oxide among V_2O_3, V_2O_4 and V_2O_5, upon reaction with alkali leads to formation of an oxide anion. The oxidation state of V in the oxide anion is :

(A) + 4 **(B)** + 5 **(C)** + 3 **(D)** + 7

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Q10. MCQ (Single Correct Answer)

Open solution

Identify product [A], [B] and [C] in the following reaction sequence.

$$CH_3 - C \equiv CH \xrightarrow[H_2]{Pd/C} [A] \xrightarrow[(ii)\ Zn,H_2O]{(i)\ O_3} [B] + [C]$$

(A) [A]: CH_3–CH=CH_2, [B]: CH_3CHO, [C]: CH_3CH_2OH

(B) [A]: CH_3–CH=CH_2, [B]: CH_3CHO, [C]: HCHO

(C) [A]: $CH_3CH_2CH_3$, [B]: CH_3CHO, [C]: HCHO

(D)

$$[A] : CH_2 = CH_2,\ [B] : H_3C - \overset{\overset{O}{\|}}{C} - CH_3,\ [C] : HCHO$$

Q11. MCQ (Single Correct Answer)

Open solution

Identify correct conversion during acidic hydrolysis from the following :

(A) starch gives galactose.

(B) cane sugar gives equal amount of glucose and fructose.

(C) milk sugar gives glucose and galactose.

(D) amylopectin gives glucose and fructose.

(E) amylose gives only glucose.

Choose the **correct** answer from the options given below :

(A) (A), (B) and (C) only

(B) (B), (C) and (D) only

(C) (B), (C) and (E) only

(D) (C), (D) and (E) only

Q12. MCQ (Single Correct Answer)

Open solution

The total number of compounds from below when treated with hot $KMnO_4$, giving benzoic acid is:



(A) 4

(B) 3

(C) 5

(D) 6

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Q13. MCQ (Single Correct Answer)

Open solution

Assume a living cell with 0.9% (w/w) of glucose solution (aqueous). This cell is immersed in another solution having equal mole fraction of glucose and water.
(Consider the data upto first decimal place only)

The cell will :

(A) shrink since solution is $0.45\%(w/w)$ as a result of association of glucose molecules (due to hydrogen bonding)

(B) shrink since solution is $0.5\%(w/w)$

(C) show no change in volume since solution is $0.9\%(w/w)$

(D) swell up since solution is $1\%(w/w)$

Q14. MCQ (Single Correct Answer)

Open solution

The purification method based on the following physical transformation is :

$$\underset{(X)}{\text{Solid}} \xrightarrow{\text{Heat}} \underset{(X)}{\text{Vapour}} \xrightarrow{\text{Cool}} \underset{(X)}{\text{Solid}}$$

(A) Extraction **(B)** Crystallization **(C)** Distillation **(D)** Sublimation

Q15. MCQ (Single Correct Answer)

Open solution

Identify correct statements :

(A) Primary amines do not give diazonium salts when treated with $NaNO_2$ in acidic condition.

(B) Aliphatic and aromatic primary amines on heating with $CHCl_3$ and ethanolic KOH form carbylamines.

(C) Secondary and tertiary amines also give carbylamine test.

(D) Benzenesulfonyl chloride is known as Hinsberg's reagent.

(E) Tertiary amines reacts with benzenesulfonyl chloride very easily.

Choose the **correct** answer from the options given below :

(A) (B) and (C) only **(B)** (D) and (E) only **(C)** (A) and (B) only **(D)** (B) and (D) only

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Q16. MCQ (Single Correct Answer)

Open solution

Identify the inorganic sulphides that are yellow in colour :

(A) $(NH_4)_2S$

(B) PbS

(C) CuS

(D) As_2S_3

(E) As_2S_5

Choose the **correct** answer from the options given below :

(A) (D) and (E) only

(B) (A) and (B) only

(C) (A), (D) and (E) only

(D) (A) and (C) only

Q17. MCQ (Single Correct Answer)

Open solution

Consider an elementary reaction

$\mathrm{A(g) + B(g) \rightarrow C(g) + D(g)}$

If the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become 'x' times of the original reaction rate. The value of x is :

(A) 3 (B) 9 (C) $\frac{1}{3}$ (D) $\frac{1}{9}$

Q18. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : According to the Law of Octaves, the elements were arranged in the increasing order of their atomic number.

Statement II : Meyer observed a periodically repeated pattern upon plotting physical properties of certain elements against their respective atomic numbers.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both **Statement I** and **Statement II** are false

(B) **Statement I** is false but **Statement II** is true

(C) **Statement I** is true but **Statement II** is false

(D) Both **Statement I** and **Statement II** are true

Q19. MCQ (Single Correct Answer)

Open solution

Concentrated nitric acid is labelled as 75% by mass. The volume in mL of the solution which contains 30 g of nitric acid is _____.

Given : Density of nitric acid solution is $1.25\ \mathrm{g/mL}$.

(A) 32 (B) 40 (C) 55 (D) 45

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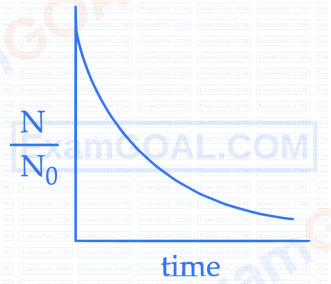
Q20. MCQ (Single Correct Answer)

For bacterial growth in a cell culture, growth law is very similar to the law of radioactive decay. Which of the following graphs is most suitable to represent bacterial colony growth ?

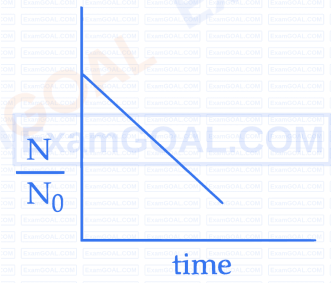
Open solution

Where N - Number of Bacteria at any time, N_0 - Initial number of Bacteria.

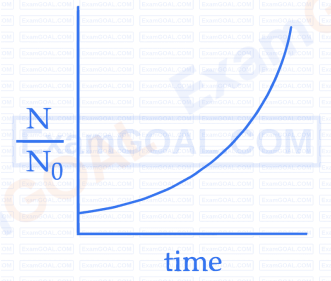
(A)



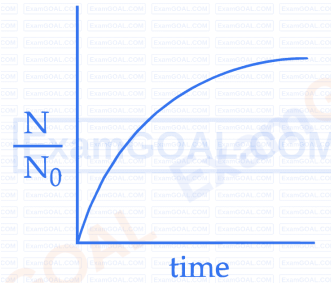
(B)



(C)



(D)



Q21. Numerical

The spin only magnetic moment (μ) value (B.M.) of the compound with strongest oxidising power among Mn_2O_3, TiO and VO is _____ B.M. (Nearest integer).

Open solution

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Q22. Numerical

Total number of molecules/species from following which will be paramagnetic is _____.

$O_2, O_2^+, O_2^-, NO, NO_2, CO, K_2[NiCl_4], [Co(NH_3)_6]Cl_3, K_2[Ni(CN)_4]$

Open solution

Q23. Numerical

Electrolysis of 600 mL aqueous solution of NaCl for 5 min changes the pH of the solution to 12 .

The current in Amperes used for the given electrolysis is _____ . (Nearest integer).

Open solution

Q24. Numerical

Consider the following data :

Heat of formation of $CO_2(g) = -393.5\ kJ\ mol^{-1}$

Heat of formation of $H_2O(l) = -286.0\ kJ\ mol^{-1}$

Heat of combustion of benzene $= -3267.0\ kJ\ mol^{-1}$

The heat of formation of benzene is _____ $kJmol^{-1}$. (Nearest integer)

Open solution

Q25. Numerical

A group 15 element forms $d\pi - d\pi$ bond with transition metals. It also forms hydride, which is a strongest base among the hydrides of other group members that form $d\pi - d\pi$ bond.

The atomic number of the element is _____ .

Open solution

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

The area of the region bounded by the curves $x(1+y^2) = 1$ and $y^2 = 2x$ is:

(A) $\frac{\pi}{4} - \frac{1}{3}$ (B) $\frac{\pi}{2} - \frac{1}{3}$

(C) $2\left(\frac{\pi}{2} - \frac{1}{3}\right)$ (D) $\frac{1}{2}\left(\frac{\pi}{2} - \frac{1}{3}\right)$

Open solution

Q27. MCQ (Single Correct Answer)

Bag B_1 contains 6 white and 4 blue balls, Bag B_2 contains 4 white and 6 blue balls, and Bag B_3 contains 5 white and 5 blue balls. One of the bags is selected at random and a ball is drawn from it. If the ball is white, then the probability that the ball is drawn from Bag B_2 is:

(A) $\frac{2}{5}$ (B) $\frac{4}{15}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Open solution

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Q28. MCQ (Single Correct Answer)

Let $\mathrm{f}: \mathrm{R} \rightarrow \mathrm{R}$ be a twice differentiable function such that $f(2)=1$. If $\mathrm{F}(\mathrm{x})=\mathrm{x} f(\mathrm{x})$ for all $\mathrm{x} \in \mathrm{R}$, $\int_0^2 xF'(x)dx = 6$ and $\int_0^2 x^2F''(x)dx = 40$, then $F'(2)+\int_0^2 F(x)dx$ is equal to :

Open solution

(A) 13 (B) 11 (C) 9 (D) 15

Q29. MCQ (Single Correct Answer)

If the midpoint of a chord of the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is $(\sqrt{2}, 4/3)$, and the length of the chord is $\frac{2\sqrt{\alpha}}{3}$, then α is :

Open solution

(A) 26 (B) 18 (C) 22 (D) 20

Q30. MCQ (Single Correct Answer)

The square of the distance of the point $\left(\frac{15}{7}, \frac{32}{7}, 7\right)$ from the line $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ in the direction of the vector $\hat{i}+4\hat{j}+7\hat{k}$ is:

Open solution

(A) 66 (B) 54 (C) 41 (D) 44

Q31. MCQ (Single Correct Answer)

Two equal sides of an isosceles triangle are along $-x+2y=4$ and $x+y=4$. If m is the slope of its third side, then the sum, of all possible distinct values of m, is:

Open solution

(A) $-2\sqrt{10}$ (B) 12 (C) -6 (D) 6

Q32. MCQ (Single Correct Answer)

Let the coefficients of three consecutive terms T_r, T_{r+1} and T_{r+2} in the binomial expansion of $(a+b)^{12}$ be in a G.P. and let p be the number of all possible values of r. Let q be the sum of all rational terms in the binomial expansion of $(\sqrt[4]{3}+\sqrt[3]{4})^{12}$. Then $p+q$ is equal to:

Open solution

(A) 295 (B) 283 (C) 299 (D) 287

Q33. MCQ (Single Correct Answer)

If the components of $\vec{a}=\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ along and perpendicular to $\vec{b}=3\hat{i}+\hat{j}-\hat{k}$ respectively, are $\frac{16}{11}(3\hat{i}+\hat{j}-\hat{k})$ and $\frac{1}{11}(-4\hat{i}-5\hat{j}-17\hat{k})$, then $\alpha^2+\beta^2+\gamma^2$ is equal to :

Open solution

(A) 16 (B) 23 (C) 26 (D) 18

Q34. MCQ (Single Correct Answer)

Let $f: \mathbf{R}-\{0\} \rightarrow (-\infty, 1)$ be a polynomial of degree 2 , satisfying $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$. If $f(\mathrm{K})=-2\,\mathrm{K}$, then the sum of squares of all possible values of K is :

Open solution

(A) 9 (B) 1 (C) 6 (D) 7

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Q35. MCQ (Single Correct Answer)

Let [x] denote the greatest integer less than or equal to x. Then the domain of $f(x) = \sec^{-1}(2[x] + 1)$ is:

(A) $(-\infty, \infty)$ (B) $(-\infty, \infty) - \{0\}$ (C) $(-\infty, -1] \cup [0, \infty)$ (D) $(-\infty, -1] \cup [1, \infty)$

Open solution

Q36. MCQ (Single Correct Answer)

Let S be the set of all the words that can be formed by arranging all the letters of the word GARDEN. From the set S, one word is selected at random. The probability that the selected word will NOT have vowels in alphabetical order is:

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Open solution

Q37. MCQ (Single Correct Answer)

If A and B are the points of intersection of the circle $x^2 + y^2 - 8x = 0$ and the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ and a point P moves on the line $2x - 3y + 4 = 0$, then the centroid of ΔPAB lies on the line :

(A) $x + 9y = 36$ (B) $9x - 9y = 32$ (C) $4x - 9y = 12$ (D) $6x - 9y = 20$

Open solution

Q38. MCQ (Single Correct Answer)

Let A, B, C be three points in xy-plane, whose position vectors are given by $\sqrt{3}\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\hat{j}$ and $a\hat{i} + (1-a)\hat{j}$ respectively with respect to the origin O . If the distance of the point C from the line bisecting the angle between the vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ is $\frac{9}{\sqrt{2}}$, then the sum of all the possible values of a is :

(A) 2 (B) 0 (C) $\frac{9}{2}$ (D) 1

Open solution

Q39. MCQ (Single Correct Answer)

If $\alpha + i\beta$ and $\gamma + i\delta$ are the roots of $x^2 - (3 - 2i)x - (2i - 2) = 0, i = \sqrt{-1}$, then $\alpha\gamma + \beta\delta$ is equal to:

(A) 2 (B) -6 (C) 6 (D) -2

Open solution

Q40. MCQ (Single Correct Answer)

If $\sum_{r=1}^{13} \left\{ \frac{1}{\sin\left(\frac{\pi}{4} + (r-1)\frac{\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{r\pi}{6}\right)} \right\} = a\sqrt{3} + b, a, b \in Z$, then $a^2 + b^2$ is equal to :

(A) 10 (B) 4 (C) 8 (D) 2

Open solution

Q41. MCQ (Single Correct Answer)

If $f(x) = \int \frac{1}{x^{1/4}(1+x^{1/4})} dx, f(0) = -6$, then $f(1)$ is equal to :

(A) $4(\log_e 2 - 2)$ (B) $\log_{e^2} 2 + 2$

(C) $2 - \log e^2$ (D) $4(\log_e 2 + 2)$

Open solution

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Q42. MCQ (Single Correct Answer)

Let $A=\begin{bmatrix}\frac{1}{\sqrt{2}} & -2\\ 0 & 1\end{bmatrix}$ and $P=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}, \theta>0$. If $B=PAP^{\top}, C=P^{\top}B^{10}P$ and the sum of the diagonal elements of C is $\frac{m}{n}$, where $\gcd(m,n)=1$, then $m+n$ is :

(A) 127 (B) 2049 (C) 258 (D) 65

Open solution

Q43. MCQ (Single Correct Answer)

Let f be a real valued continuous function defined on the positive real axis such that $g(x)=\int_0^x tf(t)dt$. If $g\left(x^3\right)=x^6+x^7$, then value of $\sum_{r=1}^{15} f\left(r^3\right)$ is :

(A) 270 (B) 340 (C) 310 (D) 320

Open solution

Q44. MCQ (Single Correct Answer)

Let $f:[0,3]\to$ A be defined by $f(x)=2x^3-15x^2+36x+7$ and $g:[0,\infty)\to B$ be defined by $g(x)=\frac{x^{2025}}{x^{2025}+1}$, If both the functions are onto and $S=\{x\in Z; x\in A \text{ or } x\in B\}$, then $n(S)$ is equal to :

(A) 29 (B) 31 (C) 30 (D) 36

Open solution

Q45. MCQ (Single Correct Answer)

For positive integers n, if $4a_n=\left(n^2+5n+6\right)$ and $S_n=\sum_{k=1}^{n}\left(\frac{1}{a_k}\right)$, then the value of $507S_{2025}$ is :

(A) 540 (B) 675 (C) 1350 (D) 135

Open solution

Q46. Numerical

Let $f(x)=\lim_{n\to\infty}\sum_{r=0}^{n}\left(\frac{\tan\left(x/2^{r+1}\right)+\tan^3\left(x/2^{r+1}\right)}{1-\tan^2\left(x/2^{r+1}\right)}\right)$. Then $\lim_{x\to 0}\frac{e^x-e^{f(x)}}{(x-f(x))}$ is equal to _____.

Open solution

Q47. Numerical

Let A and B be the two points of intersection of the line $y+5=0$ and the mirror image of the parabola $y^2=4x$ with respect to the line $x+y+4=0$. If d denotes the distance between A and B, and a denotes the area of $\triangle SAB$, where S is the focus of the parabola $y^2=4x$, then the value of $(a+d)$ is _____.

Open solution

Q48. Numerical

If $y=y(x)$ is the solution of the differential equation, $\sqrt{4-x^2}\frac{dy}{dx}=\left(\left(\sin^{-1}\left(\frac{x}{2}\right)\right)^2-y\right)\sin^{-1}\left(\frac{x}{2}\right), -2\le x\le 2, y(2)=\frac{\pi^2-8}{4}$, then $y^2(0)$ is equal to _____.

Open solution

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Q49. Numerical

The interior angles of a polygon with n sides, are in an A.P. with common difference 6°. If the largest interior angle of the polygon is 219°, then n is equal to _____.

Open solution

Q50. Numerical

The number of natural numbers, between 212 and 999, such that the sum of their digits is 15, is _____.

Open solution

Physics 25 questions

Q51. MCQ (Single Correct Answer)

A concave mirror produces an image of an object such that the distance between the object and image is 20 cm. If the magnification of the image is −3, then the magnitude of the radius of curvature of the mirror is :

(A) 3.75 cm (B) 15 cm (C) 7.5 cm (D) 30 cm

Open solution

Q52. MCQ (Single Correct Answer)

A uniform rod of mass 250 g having length 100 cm is balanced on a sharp edge at 40 cm mark. A mass of 400 g is suspended at 10 cm mark. To maintain the balance of the rod, the mass to be suspended at 90 cm mark, is

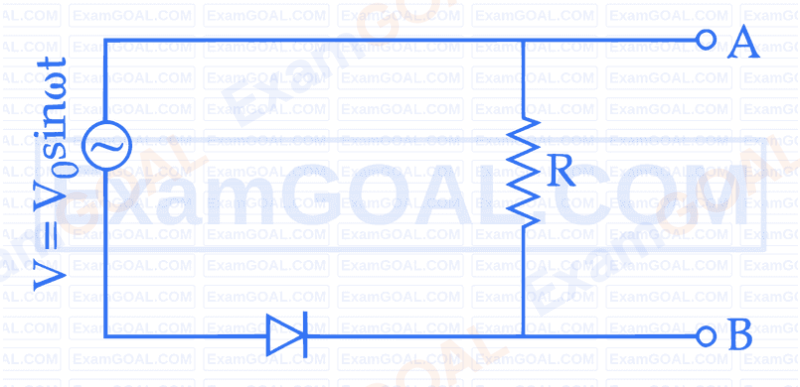
(A) 290 g (B) 200 g (C) 190 g (D) 300 g

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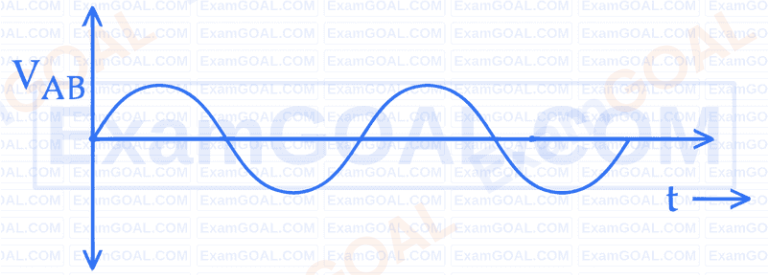
Q53. MCQ (Single Correct Answer)

Open solution



In the circuit shown here, assuming threshold voltage of diode is negligibly small, then voltage V_{AB} is correctly represented by:

(A)



(B)



(C) V_{AB} would be zero at all times.

(D)



Q54. MCQ (Single Correct Answer)

Open solution

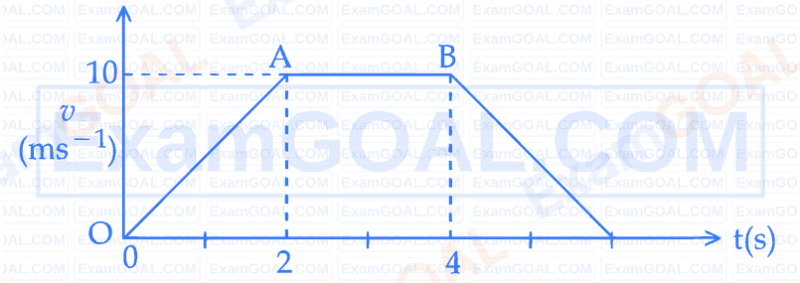
A uniform magnetic field of 0.4 T acts perpendicular to a circular copper disc 20 cm in radius. The disc is having a uniform angular velocity of 10 π rad s^{-1} about an axis through its centre and perpendicular to the disc. What is the potential difference developed between the axis of the disc and the rim? $(\pi = 3.14)$

(A) 0.5024 V (B) 0.0628 V (C) 0.2512 V (D) 0.1256 V

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Q55. MCQ (Single Correct Answer)

The velocity-time graph of an object moving along a straight line is shown in the figure. What is the distance covered by the object between $t = 0$ to $t = 4s$?



Open solution

(A) 11 m (B) 30 m (C) 13 m (D) 10 m

Q56. MCQ (Single Correct Answer)

The magnetic field of an E.M. wave is given by $\vec{B} = \left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right)30\sin\left[\omega\left(t - \frac{z}{c}\right)\right]$ (S.I. Units).

The corresponding electric field in S.I. units is:

(A) $\overrightarrow{E} = \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)30c\sin\left[\omega\left(t + \frac{z}{c}\right)\right]$

(B) $\overrightarrow{E} = \left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)30c\sin\left[\omega\left(t - \frac{z}{c}\right)\right]$

(C) $\overrightarrow{E} = \left(\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}\right)30c\sin\left[\omega\left(t + \frac{z}{c}\right)\right]$

(D) $\overrightarrow{E} = \left(\frac{3}{4}\hat{i} + \frac{1}{4}\hat{j}\right)30c\cos\left[\omega\left(t - \frac{z}{c}\right)\right]$

Open solution

Q57. MCQ (Single Correct Answer)

A balloon and its content having mass M is moving up with an acceleration 'a'. The mass that must be released from the content so that the balloon starts moving up with an acceleration '3a' will be

(Take 'g' as acceleration due to gravity)

(A) $\frac{3Ma}{2a+g}$ (B) $\frac{2Ma}{3a+g}$ (C) $\frac{3Ma}{2a-g}$ (D) $\frac{2Ma}{3a-g}$

Open solution

Q58. MCQ (Single Correct Answer)

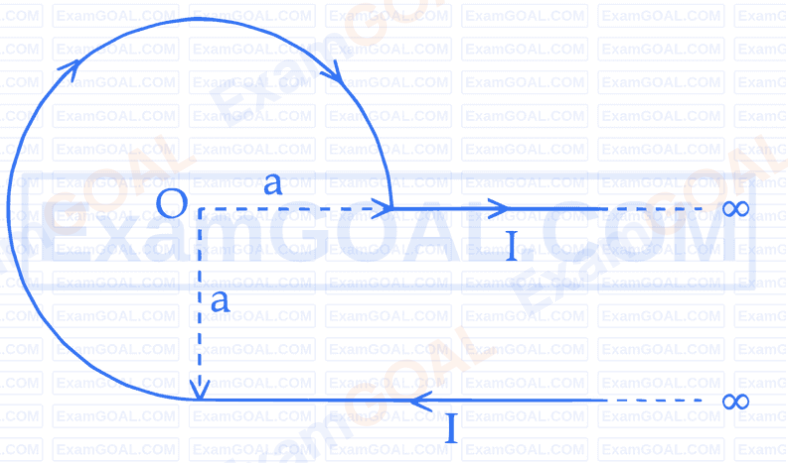
The ratio of vapour densities of two gases at the same temperature is $\frac{4}{25}$, then the ratio of r.m.s. velocities will be :

(A) $\frac{5}{2}$ (B) $\frac{25}{4}$ (C) $\frac{4}{25}$ (D) $\frac{2}{5}$

Open solution

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Q59. MCQ (Single Correct Answer)



Open solution

An infinite wire has a circular bend of radius *a*, and carrying a current *I* as shown in the figure. The magnitude of magnetic field at the origin O of the arc is given by:

(A) $\frac{\mu_0}{2\pi}\frac{I}{a}\left[\frac{\pi}{2}+2\right]$ (B) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+1\right]$

(C) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{\pi}{2}+1\right]$ (D) $\frac{\mu_0}{4\pi}\frac{I}{a}\left[\frac{3\pi}{2}+2\right]$

Q60. MCQ (Single Correct Answer)

The frequency of revolution of the electron in Bohr's orbit varies with *n*, the principal quantum number as:

(A) $\frac{1}{n^4}$ (B) $\frac{1}{n^2}$ (C) $\frac{1}{n^3}$ (D) $\frac{1}{n}$

Open solution

Q61. MCQ (Single Correct Answer)

Earth has mass 8 times and radius 2 times that of a planet. If the escape velocity from the earth is 11.2 km/s, the escape velocity in km/s from the planet will be:

(A) 8.4 (B) 11.2 (C) 5.6 (D) 2.8

Open solution

Q62. MCQ (Single Correct Answer)

Which of the following phenomena cannot be explained by wave theory of light?

(A) Refraction of light (B) Reflection of light

(C) Diffraction of light (D) Compton effect

Open solution

Q63. MCQ (Single Correct Answer)

A body of mass 4 kg is placed on a plane at a point P having coordinates $(3,4)\mathrm{m}$. Under the action of force $\vec{F}=(2\hat{i}+3\hat{j})\mathrm{N}$, it moves to a new point Q having coordinates $(6,10)\mathrm{m}$ in 4 sec . The average power and instantaneous power at the end of 4 sec are in the ratio of :

(A) 4 : 3 (B) 13 : 6 (C) 1 : 2 (D) 6 : 13

Open solution

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Q64. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II.

List - I	List - II
(A) Angular Impulse	(I) $M^0 L^2 T^{-2}$
(B) Latent Heat	(II) $M L^2 T^{-3} A^{-1}$
(C) Electrical resistivity	(III) $M L^2 T^{-1}$
(D) Electromotive force	(IV) $M L^3 T^{-3} A^{-2}$

Choose the correct answer from the options given below:

(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

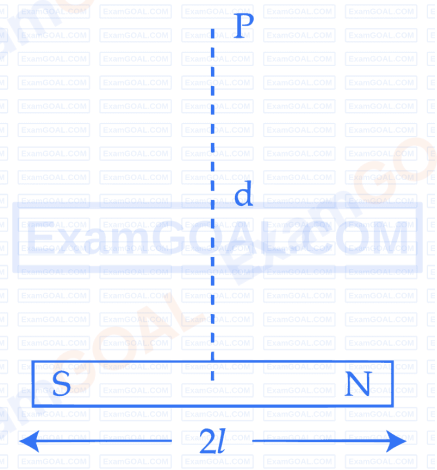
(B) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(D) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Q65. MCQ (More than One Correct Answer)

Open solution



A bar magnet has total length $2l = 20$ units and the field point P is at a distance $d = 10$ units from the centre of the magnet. If the relative uncertainty of length measurement is 1%, then uncertainty of the magnetic field at point P is :

(A) 10%

(B) 5%

(C) 3%

(D) 4%

Q66. MCQ (Single Correct Answer)

Open solution

In a long glass tube, mixture of two liquids A and B with refractive indices 1.3 and 1.4 respectively, forms a convex refractive meniscus towards A . If an object placed at 13 cm from the vertex of the meniscus in A forms an image with a magnification of -2, then the radius of curvature of meniscus is :

(A) $\frac{2}{3}$ cm

(B) $\frac{4}{3}$ cm

(C) $\frac{1}{3}$ cm

(D) 1 cm

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Q67. MCQ (Single Correct Answer)

The kinetic energy of translation of the molecules in 50 g of CO_2 gas at 17°C is :

(A) 4205.5 J (B) 3582.7 J (C) 3986.3 J (D) 4102.8 J

Open solution

Q68. MCQ (Single Correct Answer)

A 400 g solid cube having an edge of length 10 cm floats in water. How much volume of the cube is outside the water?

(Given: density of water = 1000 kg m^{-3})

(A) 400 cm^3 (B) 600 cm^3 (C) 1400 cm^3 (D) 4000 cm^3

Open solution

Q69. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Knowing initial position x_0 and initial momentum p_0 is enough to determine the position and momentum at any time t for a simple harmonic motion with a given angular frequency ω.

Reason (R) : The amplitude and phase can be expressed in terms of x_0 and p_0.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

Open solution

Q70. MCQ (Single Correct Answer)

A parallel plate capacitor of capacitance 1 μF is charged to a potential difference of 20 V. The distance between plates is 1 μm. The energy density between plates of capacitor is :

(A) 1.8×10^3 J/m^3 (B) 2×10^2 J/m^3 (C) 2×10^{-4} J/m^3 (D) 1.8×10^5 J/m^3

Open solution

Q71. Numerical

A thin transparent film with refractive index 1.4 , is held on circular ring of radius 1.8 cm . The fluid in the film evaporates such that transmission through the film at wavelength 560 nm goes to a minimum every 12 seconds. Assuming that the film is flat on its two sides, the rate of evaporation is _____ $\pi \times 10^{-13}\ m^3/s$.

Open solution

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Q72. Numerical

The volume contraction of a solid copper cube of edge length 10 cm , when subjected to a hydraulic pressure of 7×10^6 Pa, would be _____ mm^3.

(Given bulk modulus of copper $= 1.4 \times 10^{11}\ N\,m^{-2}$)

Open solution

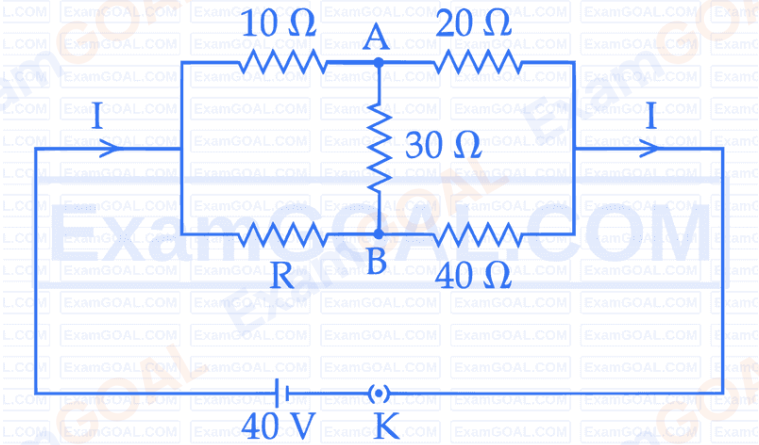
Q73. Numerical

An electric dipole of dipole moment $6 \times 10^{-6} Cm$ is placed in uniform electric field of magnitude $10^6 V/m$. Initially, the dipole moment is parallel to electric field. The work that needs to be done on the dipole to make its dipole moment opposite to the field, will be _____ J.

Open solution

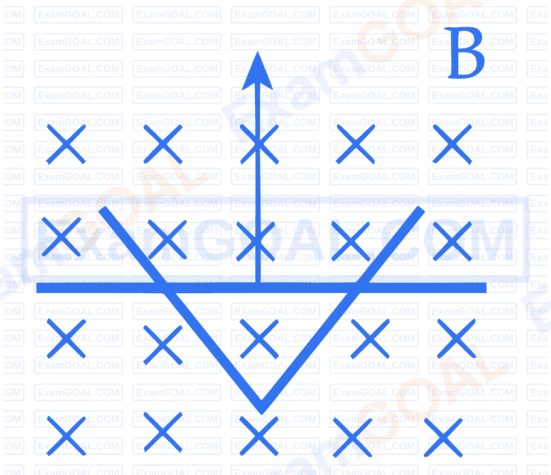
Q74. Numerical

The value of current I in the electrical circuit as given below, when potential at A is equal to the potential at B, will be _____ A.



Open solution

Q75. Numerical



A conducting bar moves on two conducting rails as shown in the figure. A constant magnetic field B exists into the page. The bar starts to move from the vertex at time t = 0 with a constant velocity. If the induced EMF is $E \propto t^n$, then value of n is _____.

Open solution

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Answer Key

Chemistry

1. **D**	2. **C**	3. **B**	4. **D**	5. **C**
6. **A**	7. **C**	8. **D**	9. **B**	10. **B**
11. **C**	12. **C**	13. *Scan QR*	14. **D**	15. **D**
16. **A**	17. **B**	18. **A**	19. **A**	20. **C**
21. **5**	22. **6**	23. **2**	24. **48**	25. **15**

Mathematics

26. **B**	27. **B**	28. **B**	29. **C**	30. **A**
31. **D**	32. **B**	33. **C**	34. **C**	35. **A**
36. **B**	37. **D**	38. **D**	39. **A**	40. **C**
41. **A**	42. **D**	43. **C**	44. **C**	45. **B**
46. **1**	47. **14**	48. **4**	49. **20**	50. **64**

Physics

51. **B**	52. **C**	53. **B**	54. **C**	55. **B**
56. **B**	57. **B**	58. **A**	59. **B**	60. **C**
61. **C**	62. **D**	63. **D**	64. **C**	65. **C, D**
66. **A**	67. **D**	68. **B**	69. **D**	70. **A**
71. **54**	72. **50**	73. **12**	74. **2**	75. **1**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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